

**NMAM INSTITUTE OF TECHNOLOGY, NITTE***(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (AIML/CCE) (Credit System) Degree Examinations**

May - June 2023

**20AM621 / 20CC621 - BLOCKCHAIN TECHNOLOGY**

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

**Unit – I**

|                                                                                                                                           | Marks | BT* | CO* | PO* |
|-------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1. a) Explain various ideas that supported the invention of Bit coin and block chain with a neat diagram.                                 | 10    | L*2 | 1   | 1   |
| b) Explain any five types of Consensus mechanism.                                                                                         | 10    | L2  | 2   | 1   |
| 2. a) Define Blockchain. What are the Generic Elements of a block chain? Explain briefly.                                                 | 10    | L2  | 1   | 1   |
| b) Differentiate between Stream cipher and Block cipher.                                                                                  | 10    | L2  | 2   | 1   |
| 3. a) Describe the working of the DAO, DAC, DAS and DApps decentralized organizations.                                                    | 10    | L2  | 1   | 1   |
| b) With suitable diagram, explain how public key cryptography is applied for encryption-decryption process and Digital signature process. | 10    | L2  | 2   | 1   |

**Unit – II**

|                                                                                                                                          |    |    |   |   |
|------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|---|
| 4. a) With suitable diagram, explain the Ethereum Virtual Machine operation.                                                             | 10 | L2 | 3 | 1 |
| b) With a simple example program, explain the layout of a solidity program.                                                              | 10 | L2 | 4 | 3 |
| 5. a) Explain ANY FOUR precompiled contracts in Ethereum.                                                                                | 10 | L2 | 3 | 1 |
| b) With a neat generic model of an oracle and smart contract ecosystem, explain the role and types of oracle in smart contract paradigm. | 10 | L2 | 4 | 3 |
| 6. a) Explain the following:<br>i) Ethereum Stack<br>ii) Recardian Contract                                                              | 10 | L2 | 3 | 1 |
| b) With suitable example, explain following in Solidity.<br>i) Events<br>ii) Inheritance                                                 | 10 | L2 | 4 | 3 |

**Unit – III**

|                                                                                                    |    |    |   |   |
|----------------------------------------------------------------------------------------------------|----|----|---|---|
| 7. a) With a neat runtime architecture of Hyperledger fabric, explain transaction processing flow. | 10 | L2 | 5 | 2 |
| b) Explain the principles followed in designing any Hyperledger projects.                          | 10 | L2 | 5 | 2 |
| 8. a) Explain five tools in the Hyperledger project.                                               | 10 | L2 | 5 | 2 |
| b) Explain advantages of Hyperledger and its family of projects to the industry.                   | 10 | L2 | 5 | 2 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

\*\*\*\*\*